



Kinetic Theory of Decentralized Learning for Smart Active Matter

Gerhard Jung , Misaki Ozawa , and Eric Bertin

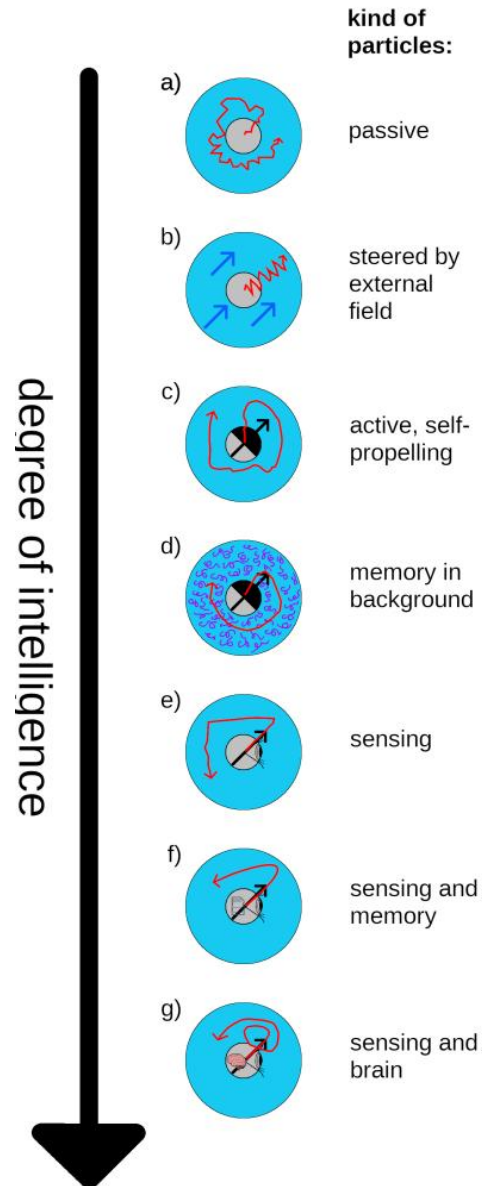
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曹溪能

内容回顾

background:smart active particle



“smart active particle” in this article is at “g” level:
sense its neighbors, memory, make and change policy

background: what is kinetic theory?

kinetic theory: solve the distribution function in phase space $\Gamma = (q_1, q_2, \dots, q_N, p_1, p_2, \dots, p_N)$ 6N dimension

microscopic

Hamilton equation

$$H = H(\Gamma)$$

$$\dot{q}_i = \{q_i, H\}, \dot{p}_i = \{p_i, H\}$$

dynamic theory

Liouville equation

$$\rho = \rho(\Gamma, t)$$

$$\partial_t \rho + \{\rho, H\} = 0$$

mesoscopic

boltzman equation

$$f_i(q, p, t) = \int dp_1 \dots dp_{i-1} dp_{i+1} \dots dp_N dq_1, \dots dq_{i-1} dq_{i+1} \dots dq_N \rho(\Gamma, t)$$

$$\partial_t f + \{f, \underbrace{H_1}_{\text{one-body}}\} = \underbrace{C[f]}_{\text{collisions}}$$

macroscopic

hydrodynamic equation

$$\rho = \int f(q, p, t) dp, \quad u = \int p f(q, p, t) dp$$

$$\partial_t \rho + \nabla \cdot (\rho \mathbf{u}) = 0$$

$$\partial_t \mathbf{u} + (\mathbf{u} \cdot \nabla) \mathbf{u} = -\frac{1}{\rho} \nabla p + \nu \nabla^2 \mathbf{u} + \mathbf{f}$$

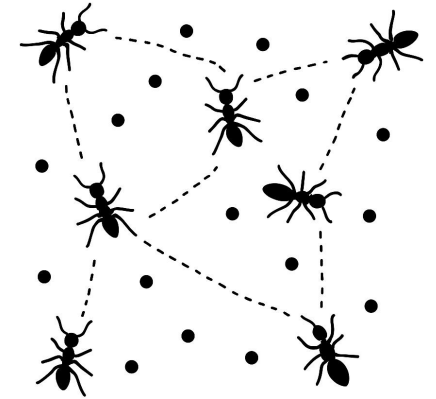
background :decentralized learning

what is learning?

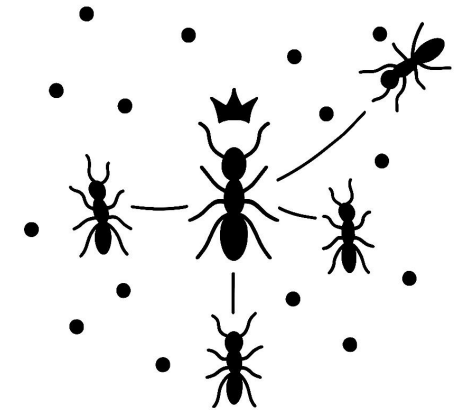
have a target and change its parameters to try to realize the target

what is decentralized learning?

for each agent,update their parameters using only local data and neighbor interactions,no central controller



decentralized learning



centralized learning

generated by gpt

background :decentralized learning

Does this paper's research belong to machine learning?

No, it's the research paradigm of kinetic theory in physics, not the paradigm of machine learning

Why do we feel that this article is related to machine learning?

One perspective: The article borrows some concepts from reinforcement learning, such as policy and reward function, and decentralized learning also exists in reinforcement learning.

More comparisons between this article and “decentralized learning Multi-Agent Reinforcement Learning” will be introduced after this article.

model:Agent-based model of decentralized learning

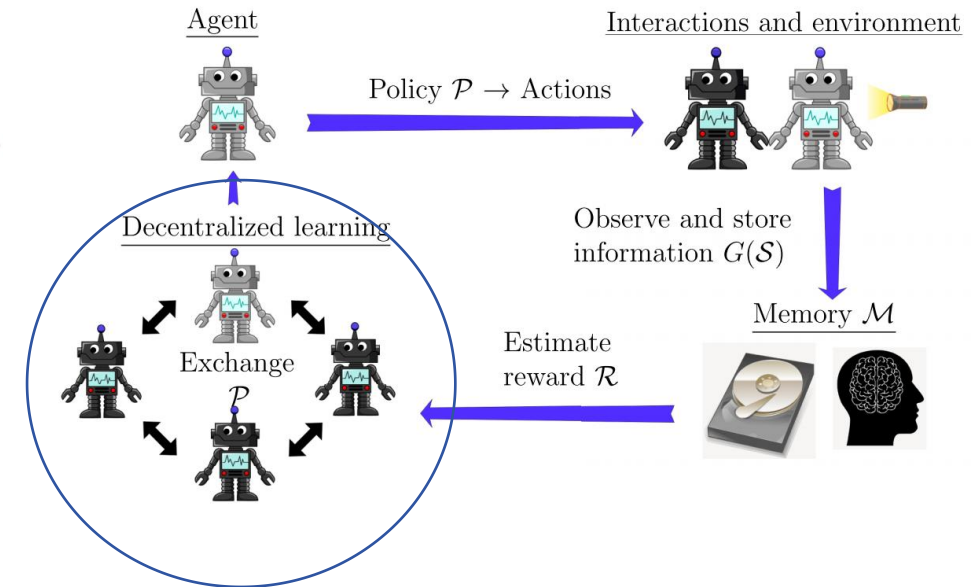
what is an agent?

- position x , overdamped velocity v , position diffusivity D
- state S : internal physical variables e.g., orientation, self-propulsion velocity
- policy P : parameters that agent can change e.g., angular diffusion coefficient
- processed information $G(s)$, and memory M : $\frac{dM(t)}{dt} = -\frac{1}{\tau_M} [M(t) - G(S(t))]$,

decentralized learning procedure:

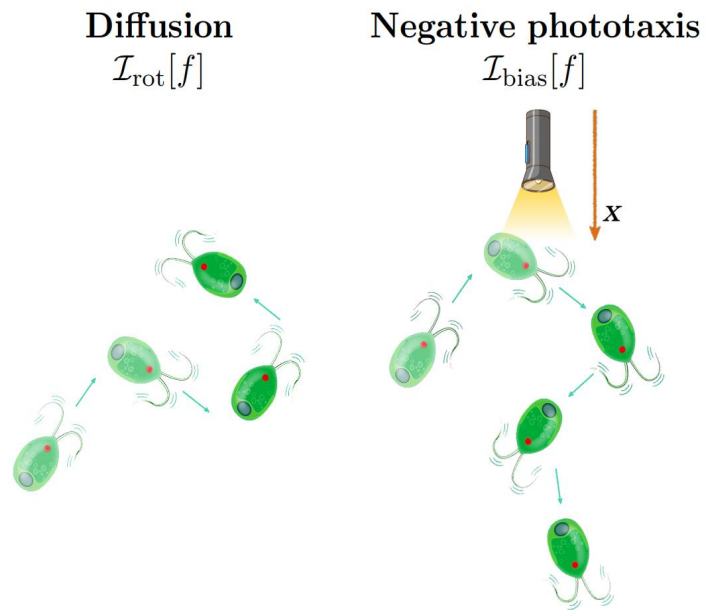
- teaching and learning at rate $\lambda_T = \tau_T^{-1}$ τ_T is the teaching time
- i become a teacher at the possibility $p_T(\mathcal{R}_i, \mathcal{R}_n) = \frac{1}{2} (1 + \tanh[\alpha_T(\mathcal{R}_i - \mathcal{R}_n)])$
- student copy teacher's policy and memory $\mathcal{P}_S^{\text{new}} = \mathcal{P}_T$ $\mathcal{M}_S^{\text{new}} = \mathcal{M}_T$
- add mutation in policy space, with diffusion coefficient D_{mut}

time scale separation $\tau_G \ll \tau_M \ll \tau_T$ (time scale of $S \ll M \ll P$)



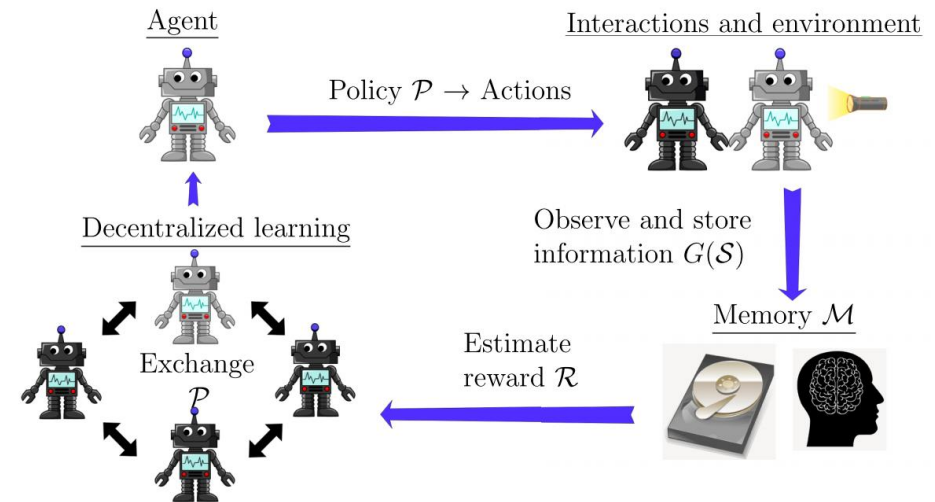
model: one example

Phototactic, diffusive agents



$$\begin{aligned}\mathcal{S} &= \theta \\ G(\mathcal{S}) &= v_0 \cos(\theta) \\ \mathcal{P} &= D_\theta\end{aligned}$$

$$\mathcal{R}(\mathcal{M}) = -(\mathcal{M} - V_T)^2 \quad V_T \text{ is constant}$$



kinetic theory

- single-agent phase-space density $f(S, \mathcal{M}, \mathcal{P}, \mathbf{x}, t)$
- equation $\frac{\partial f}{\partial t} = \mathcal{I}_{\text{phys}}[f] + \mathcal{I}_{\text{mem}}[f] + \mathcal{I}_{\text{learn}}[f]$
 - $\mathcal{I}_{\text{phys}}[f]$ physical dynamic of the state S and the position $\mathbf{x}$
 - $\mathcal{I}_{\text{mem}}[f]$ memory dynamic of M
 - $\mathcal{I}_{\text{learn}}[f]$ policy dynamic of P $\mathcal{I}_{\text{learn}}[f] = D_{\text{mut}} \frac{\partial^2 f}{\partial \mathcal{P}^2} + \mathcal{I}_{\text{teach}}[f]$
- **time scale separation** $\tau_G \ll \tau_{\mathcal{M}} \ll \tau_T$ (time scale of S $\ll$ M $\ll$ P)
- we want to get **the hydrodynamic of policy(its mean and variance)**
- Integrate over slow variables S and M $\varphi_0(\mathcal{P}, \mathbf{x}, t) = \int dS \int d\mathcal{M} f(S, \mathcal{M}, \mathcal{P}, \mathbf{x}, t)$
- Obtain P's hydrodynamic equations $\mu(\mathbf{x}, t), \sigma^2(\mathbf{x}, t)$

$$\mu(\mathbf{x}, t) = \rho^{-1} \int d\mathcal{P} \mathcal{P} \varphi_0(\mathcal{P}, \mathbf{x}, t),$$

$$\sigma^2(\mathbf{x}, t) = \rho^{-1} \int d\mathcal{P} (\mathcal{P} - \mu(\mathbf{x}, t))^2 \varphi_0(\mathcal{P}, \mathbf{x}, t)$$

$$\text{where } \rho(\mathbf{x}, t) = \int d\mathcal{P} \varphi_0(\mathcal{P}, \mathbf{x}, t)$$

kinetic theory: framework and results

- P's hydrodynamic equations $\mu(x, t), \sigma^2(x, t)$ (detailed derivation see SI)

$$\frac{d\mu}{dt} = \rho \sigma^2 F_1(\mu, \sigma^2; \mu_{\mathcal{M}}^{(n)}),$$

$$\frac{d\sigma^2}{dt} = 2D_{\text{mut}} - \rho \sigma^4 F_2(\mu, \sigma^2; \mu_{\mathcal{M}}^{(n)})$$

$F_m(\mu, \sigma^2; \mu_{\mathcal{M}}^{(n)})$ with $m = 1, 2$ are model-specific polynomials

$$\mu_{\mathcal{M}}(\mathcal{P}, \mathbf{x}, t) = \varphi_0^{-1} \int d\mathcal{S} \int d\mathcal{M} \mathcal{M} f(\mathcal{S}, \mathcal{M}, \mathcal{P}, \mathbf{x}, t)$$

$$\mu_{\mathcal{M}}(\mathcal{P}, \mathbf{x}, t) = \mu_{\mathcal{M}}^{(0)}(\mathbf{x}, t) + \mu_{\mathcal{M}}^{(1)}(\mathbf{x}, t)(\mathcal{P} - \mathcal{P}_*) + \dots$$

- remarks and nontrivial conclusions:

1. “physical” parameters lie in $\mu_{\mathcal{M}}^{(n)}$, “memory” parameters lie in F_m , because of timescale separation
2. $\dot{\mu} \propto \rho$ the higher the density, the faster the evolution.
3. $\dot{\mu} \propto \sigma^2$ consistent with fisher's fundamental theorem of natural selection
4. on timescale $t \gg \tau_m$, we have $F_2 > 0$. the variance will decay to 0 in the absence of D_{mut} . D_{mut} act as driven force, break detailed balance and maintain the diversity

example: Phototactic, diffusive agents

- target $\langle V_x \rangle = N^{-1} \sum_{i=1}^N V_{x,i}$ equal to V_T

- hydrodynamic equation
$$\sigma^2(t) = \sqrt{\frac{2D_{\text{mut}}}{\lambda_0}} \frac{1}{\tanh[\sqrt{2D_{\text{mut}}\lambda_0}(\tau_0 + t)]}$$

$$\mu(t) = D_{\theta,T} + \frac{\eta_0 \sqrt{2D_{\text{mut}}\lambda_0}}{\sinh[\sqrt{2D_{\text{mut}}\lambda_0}(\tau_0 + t)]},$$

- characteristic learning rate $\lambda_0 = 4\lambda_T\alpha_T\rho V_*'^2$

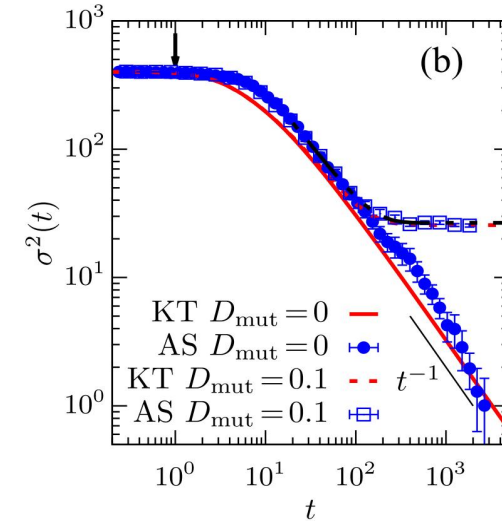
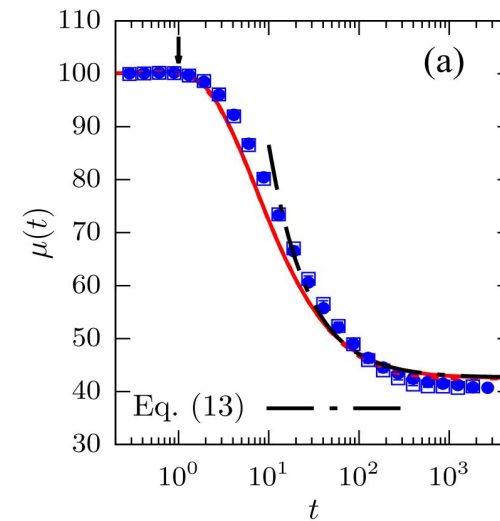
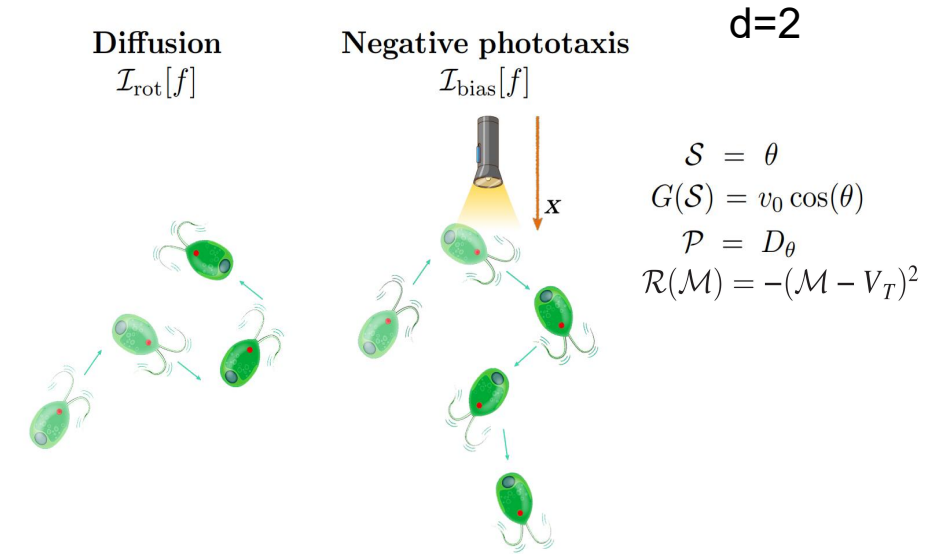
steady policy invariance $\sigma_\infty^2 = \sqrt{2D_{\text{mut}}/\lambda_0}$

learning time $\tau_L = (2D_{\text{mut}}\lambda_0)^{-1/2}$

→ **uncertainty relation** $\tau_L \sigma_\infty^2 = \lambda_0^{-1}$

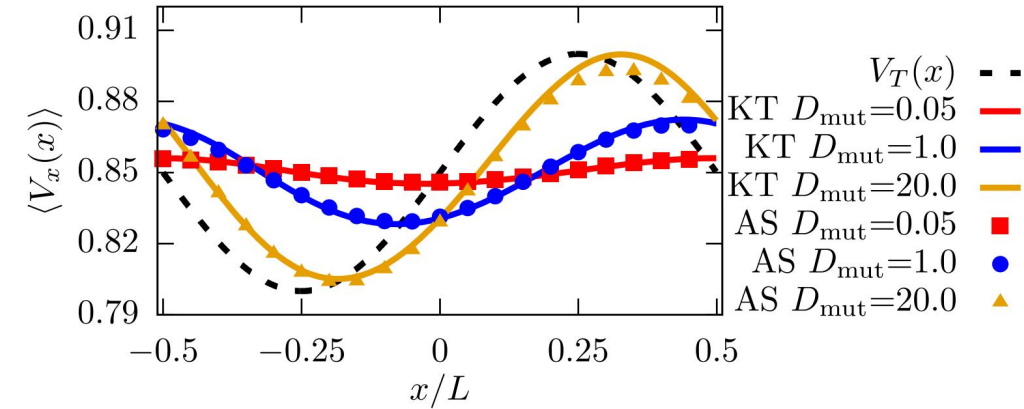
quick learn is at the cost of large fluctuation around the target policy

or you cannot learn quickly and well at the same learning rate



example: Phototactic, diffusive agents

- target $V_T(x) = V_T^s + V_T^\Delta \sin(2\pi x/L)$
- steady velocity $\langle V_x(x) \rangle = V_T^s + V_x^\Delta(\phi) \sin(2\pi x/L - \phi)$
 $\tan(\phi) = 2\pi V_T^s / (\sqrt{2D_{\text{mut}}}\lambda_0 L)$
 $V_x^\Delta(\phi) = V_T^\Delta \cos(\phi)$

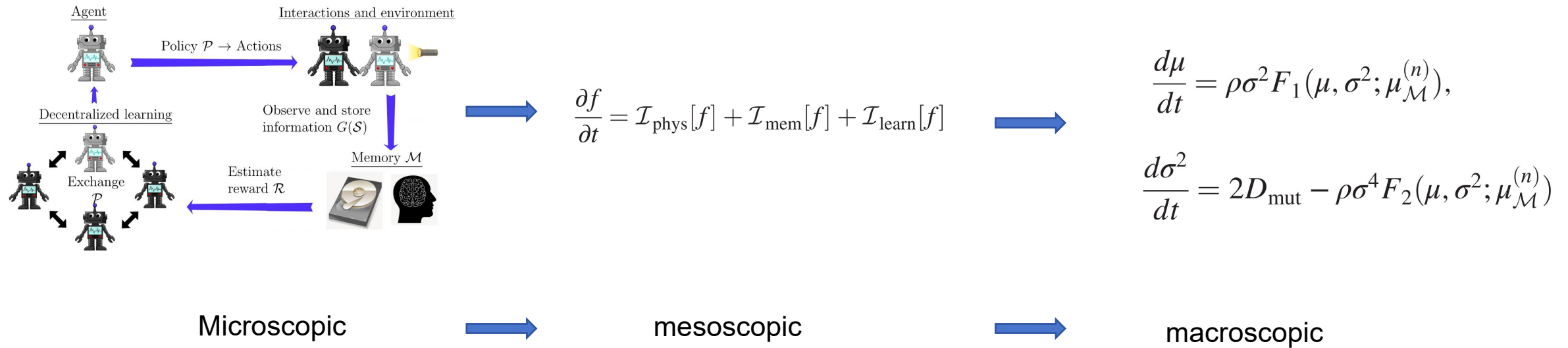


Agents with small mutation rates learn too slowly to adapt in time to the environment.

does it machine learning?no
what's the relationship?

conclusion

- kinetic theory framework for decentralized learning in smart active matter
based on the assumption of **timescale separation**



- uncertainty relation** between learning time and policy fluctuations. $\tau_L \sigma_{\infty}^2 = \lambda_0^{-1}$

“you cannot learn quickly and well at the same learning rate”

Appendix:kinetic theory

$$\frac{\partial f}{\partial t} = \mathcal{I}_{\text{phys}}[f] + \mathcal{I}_{\text{mem}}[f] + \mathcal{I}_{\text{learn}}[f]$$

- derive explicit expressions from microdynamic rules

$$\mathcal{I}_{\text{phys}}[f]$$

$$\mathcal{I}_{\text{mem}}[f] \quad \text{Continuity Equation} \quad \frac{\partial f}{\partial t} + \frac{\partial}{\partial M}(\dot{M}f) = 0$$

$$\text{dynamic rule} \quad \frac{d\mathcal{M}(t)}{dt} = -\frac{1}{\tau_{\mathcal{M}}}[\mathcal{M}(t) - G(S(t))],$$

$$\begin{aligned} \longrightarrow \quad \mathcal{I}_{\text{mem}}[f] &= -\frac{\partial}{\partial M} \left[-\frac{1}{\tau_M} (M - G(S)) \cdot f \right] \\ &= \tau_M^{-1} \left(f + (M - G(S)) \frac{\partial f}{\partial M} \right) \end{aligned}$$

Appendix:kinetic theory

$$\mathcal{I}_{\text{teach}}[f]$$

transition rate

$$W_f(\mathcal{M}', \mathcal{P}' | \mathcal{M}, \mathcal{P}; \mathbf{x}, t) = 2\lambda_T \int d\mathcal{S}_2 \int d\mathcal{M}_2 \int d\mathcal{P}_2 \int d\mathbf{x}_2 K(\mathbf{x}_2, \mathbf{x}) \times p_T(\mathcal{R}(\mathcal{M}_2), \mathcal{R}(\mathcal{M})) f(\mathcal{S}_2, \mathcal{M}_2, \mathcal{P}_2, \mathbf{x}_2, t) \times \delta(\mathcal{P}' - \mathcal{P}_2) \delta(\mathcal{M}' - \mathcal{M}_2).$$

with kernel $K(\mathbf{x}_2, \mathbf{x}) = (2\pi\delta_c^2)^{-1} \exp[-(\mathbf{x}_2 - \mathbf{x})^2 / 2\delta_c^2]$



assumption:communication speed $\ll$ motion speed

$$\lambda_T \delta_c \ll v_0$$

$$W_f(\mathcal{M}', \mathcal{P}' | \mathcal{M}, \mathcal{P}) = 2\lambda_T p_T(\mathcal{R}(\mathcal{M}'), \mathcal{R}(\mathcal{M})) f_0(\mathcal{M}', \mathcal{P}'),$$

with $f_0(\mathcal{M}, \mathcal{P}) = \int d\mathcal{S} f(\mathcal{S}, \mathcal{M}, \mathcal{P})$

$$\mathcal{I}_{\text{teach}}[f] = \int d\mathcal{M}' \int d\mathcal{P}' [W_f(\mathcal{M}, \mathcal{P} | \mathcal{M}', \mathcal{P}') f(\mathcal{S}, \mathcal{M}', \mathcal{P}') - W_f(\mathcal{M}', \mathcal{P}' | \mathcal{M}, \mathcal{P}) f(\mathcal{S}, \mathcal{M}, \mathcal{P})]. \quad (\text{A4})$$

$$\begin{aligned} &= 2\lambda_T f_0(\mathcal{M}, \mathcal{P}) \int d\mathcal{M}' p_T(\mathcal{R}, \mathcal{R}') \\ &\quad \times \int d\mathcal{P}' f(\mathcal{S}, \mathcal{M}', \mathcal{P}') - 2\lambda_T f(\mathcal{S}, \mathcal{M}, \mathcal{P}) \\ &\quad \times \int d\mathcal{M}' p_T(\mathcal{R}', \mathcal{R}) \int d\mathcal{P}' f_0(\mathcal{M}', \mathcal{P}'), \\ &= 2\lambda_T f_0 \int d\mathcal{M}' \tanh(\alpha_T(\mathcal{R} - \mathcal{R}')) \times \int d\mathcal{P}' f_0(\mathcal{M}', \mathcal{P}') \end{aligned}$$

Appendix:kinetic theory

$$\frac{\partial f}{\partial t} = \mathcal{I}_{\text{phys}}[f] + \mathcal{I}_{\text{mem}}[f] + \mathcal{I}_{\text{learn}}[f]$$



$$\begin{aligned} \frac{\partial f_0(\mathcal{M}, \mathcal{P})}{\partial t} = & -\nabla J_0 + \tau_{\mathcal{M}}^{-1} \left(f_0 + (\mathcal{M} - \bar{G}(\mathcal{P})) \frac{\partial f_0}{\partial \mathcal{M}} \right) \\ & + 2\lambda_T f_0 \int d\mathcal{M}' \tanh(\alpha_T(\mathcal{R} - \mathcal{R}')) \\ & \times \int d\mathcal{P}' f_0(\mathcal{M}', \mathcal{P}') + D_{\text{mut}} \frac{\partial^2}{\partial \mathcal{P}^2} f_0, \end{aligned}$$

$$\text{with } J_0 = (\bar{V}(\mathcal{P}) - D_0 \nabla) f_0$$

define:

policy's pdf:

$$\varphi_0(\mathcal{P}, \mathbf{x}, t) = \int d\mathcal{S} \int d\mathcal{M} f(\mathcal{S}, \mathcal{M}, \mathcal{P}, \mathbf{x}, t)$$

mean memory:

$$\mu_{\mathcal{M}}(\mathcal{P}, \mathbf{x}, t) = \varphi_0^{-1} \int d\mathcal{M} \mathcal{M} f_0(\mathcal{M}, \mathcal{P}, \mathbf{x}, t)$$

density

$$\rho(\mathbf{x}, t) = \int d\mathcal{P} \varphi_0(\mathcal{P}, \mathbf{x}, t)$$

mean policy

$$\mu(\mathbf{x}, t) = \rho^{-1} \int d\mathcal{P} \mathcal{P} \varphi_0(\mathcal{P}, \mathbf{x}, t),$$

policy's invariance

$$\sigma^2(\mathbf{x}, t) = \rho^{-1} \int d\mathcal{P} (\mathcal{P} - \mu(\mathbf{x}, t))^2 \varphi_0(\mathcal{P}, \mathbf{x}, t)$$

$$\begin{aligned} \frac{\partial \varphi_0(\mathcal{P})}{\partial t} = & -\nabla J + D_{\text{mut}} \frac{\partial^2}{\partial \mathcal{P}^2} \varphi_0(\mathcal{P}) \\ & + \tilde{\lambda}_T \rho \int d\mathcal{M} f_0(\mathcal{M}, \mathcal{P}) \mathcal{R}(\mathcal{M}) \\ & - \tilde{\lambda}_T \varphi_0(\mathcal{P}) \int d\mathcal{P}' \int d\mathcal{M}' f_0(\mathcal{M}', \mathcal{P}') \mathcal{R}(\mathcal{M}') \end{aligned}$$

$$\frac{\partial \mu_{\mathcal{M}}(\mathcal{P})}{\partial t} = \tau_{\mathcal{M}}^{-1} (\bar{G}(\mathcal{P}) - \mu_{\mathcal{M}}(\mathcal{P}))$$



.... and depend on model detail

$$\frac{d\mu}{dt} = \rho \sigma^2 F_1(\mu, \sigma^2; \mu_{\mathcal{M}}^{(n)}),$$

$$\frac{d\sigma^2}{dt} = 2D_{\text{mut}} - \rho \sigma^4 F_2(\mu, \sigma^2; \mu_{\mathcal{M}}^{(n)}),$$

